8	**(a)**	A field of force is a region of space in which bodies may experience a force associated with the field. Or: A field of force is a region of space in which an object experiences a force. [A1]	
	(b)	**(i)**	Each ion will experience two forces that are equal in magnitude but opposite in direction, electric force (downwards) and magnetic force (upwards). [B1] Hence, the resultant force acting on the ion is zero that results in a straight-line motion. [B1]
		(ii)	The plate P is at a higher potential than plate Q. [A1] This forms a uniform electric field that is pointing downwards. Hence, the positive ions experience will experience a downwards electric force. [M1]
		(iii)	$F_E = F_B$ $qE = B_1qv$ $v = \frac{E}{B_1} = \frac{3.0 \times 10^4}{0.25}$ [C1] $= 1.2 \times 10^5 \text{ m s}^{-1}$ [A1]
		(iv)	From the drawing, at the point of the entry into B_2, the ion experiences an upward resultant magnetic force that points towards the centre of the circle. [M1] By Fleming's left-hand rule, since the positive ion is moving to the right, the magnetic flux density would be pointing into the plane of paper. [A1]
		(v)	As the ion moves within B_2, it continues to experience a magnetic force that is perpendicular to the direction of its motion. [B1] Therefore, since work done is given by $W = Fs\cos\theta$, there is no work done by the magnetic force. [B1]
		(vi)	Within B_2 region, Magnetic force provides for centripetal force. [B1] $B_2ev = m\frac{v^2}{r}$ [B1] $m = \frac{eB_2r}{v}$ $m = \frac{eB_1B_2r}{E}$ (since $v = \frac{E}{B_1}$) [A0]

		(vii)	$m_x = \dfrac{qB_1B_2r_x}{E}$ $= \dfrac{1.6\times10^{-19}(0.25)(0.60)(0.074)}{3.0\times10^{4}}$ $= 5.92\times10^{-26}$ kg [A1]
		(viii)	$m_x = \dfrac{qB_1B_2r_x}{E}$ $\%\dfrac{\Delta m_x}{m_x} = \%\dfrac{\Delta B_1}{B_1} + \%\dfrac{\Delta B_2}{B_2} + \%\dfrac{\Delta r_x}{r_x} + \%\dfrac{\Delta E}{E}$ $= 1+1+(\dfrac{0.001}{0.074}\times100)+2$ [C1] $= 5.4\%$ [A1]
		(ix)	$\Delta r = 0.079 - 0.074 = 0.005$ m $\sigma = 0.001$ m [B1: correct Δr and σ] Since $\Delta r > 3\sigma$, the ions X and Y are distinguishable. [B1]
		(x)	Any of the following with proper explanation [B1, B1]. Decrease B_2. This increases radius separation for different masses, thus improving the resolution. Improve vacuum. This reduces collision of the ions with gas molecules. Use more precise detection surface. This reduces uncertainty in the radius.